



Assignments

Strings – 1

Q1. Input a string of size n and update all the odd positions in the string to character '#'. Consider 0-based indexing.

Input : str = "PbwcskhuiglHds"

Output : "P#w#s#k#i#l#l#s"

Input : str = "a"

Output : "a"

Solution :

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    int n;
    cin >> n;
    string s;
    cin >> s;
    for (int i = 1; i < (int)s.size(); i += 2) {
        s[i] = '#';
    }
    cout << s;
}
```

Q2. Input a string of length n and count all the consonants in the given string.

Input : "pwians"

Output : 4

Input : "abdc"

Output : 3

Solution :

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    int n;
    cin >> n;
    string s;
    cin >> s;
    int c = 0;
    for (int i = 0; i < (int)s.size(); i++) {
```

```
if (s[i] == 'a' || s[i] == 'e' || s[i] == 'o' || s[i] == 'u' ||  
s[i] == 'i')  
{  
c++;  
}  
}  
cout << n - c;  
}
```

Q3. Check whether the given string is palindrome or not.

Input : "abcde"

Output : No

Input : "abcdcba"

Output : Yes

Solution :

```
#include <bits/stdc++.h>  
using namespace std;  
bool check(string &s) {  
int i = 0, j = (int)s.size() - 1;  
while (i <= j) {  
if (s[i] != s[j]) return false;  
i++, j--;  
}  
}  
int main() {  
int n;  
cin >> n;  
string s;  
cin >> s;  
cout << (check(s) ? "YES" : "NO");  
}
```

Q4. Input a string of even length and reverse the second half of the string.

Input : str = "abcdefgh"

Output : abcdhgfe

Input : str = "pwians"

Output : pwisna

Solution :

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    int n;
    cin >> n;
    string s;
    cin >> s;
    if(s.size()%2 != 0) cout<<"Invalid input."<<endl;
    else{
        reverse(s.begin() + n / 2, s.end());
        cout << s;
    }
}
```

Q5. Input a string of length less than 10 and convert it into integer without using builtin function.

Input : "3244"

Output : 3244

Input : "12"

Output : 12

Solution :

```
#include <bits/stdc++.h>
using namespace std;
int main() {
    string s;
    cin >> s;
    int val = 0, pw = 1;
    while (s.size()) {
        val += pw * (s.back() - '0');
        s.pop_back();
        pw *= 10;
    }
    cout << val;
}
```



**THANK
YOU !**

